whether a CoT step is correct, given the task and previous steps. Due to the limited data, we include in training the CoT steps that occur after the first mistake step. These steps are considered incorrect for the purposes of training (despite not being human-annotated as such).

D User interface

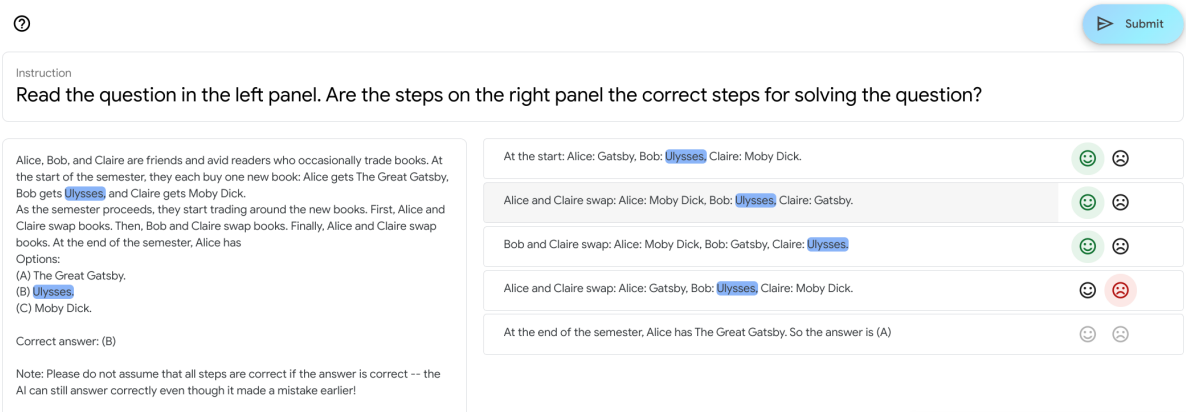

Figure 4: Screenshot of the user interface for a question from the tracking shuffled objects task.

E Benchmark scores

Model	Direct (trace)	Direct (step)	CoT (step)
Word sorting			
GPT-4-Turbo	67.74	38.24	–
GPT-4	88.24	82.35	58.82
GPT-3.5-Turbo	100.00	97.06	20.59
Gemini Pro	44.12	–	–
PaLM 2 Unicorn	100.00	73.53	35.29
Tracking shuffled objects			
GPT-4-Turbo	90.00	77.50	–
GPT-4	82.50	82.50	80.00
GPT-3.5-Turbo	67.50	0.00	0.00
Gemini Pro	12.50	–	–
PaLM 2 Unicorn	100.00	85.00	47.50
Logical deduction			
GPT-4-Turbo	100.00	83.33	–
GPT-4	100.00	100.00	0.00
GPT-3.5-Turbo	100.00	50.00	100.00
Gemini Pro	33.33	–	–
PaLM 2 Unicorn	100.00	100.00	50.00
Multistep arithmetic			
GPT-4-Turbo	57.69	40.32	–
GPT-4	53.23	46.77	27.42
GPT-3.5-Turbo	96.77	79.03	58.06
Gemini Pro	83.87	–	–
PaLM 2 Unicorn	83.87	93.55	29.03
Dyck languages			
GPT-4-Turbo	96.42	30.00	–
GPT-4	98.41	78.57	13.79
GPT-3.5-Turbo	95.74	4.76	0.00
Gemini Pro	0.00	–	–
PaLM 2 Unicorn	100.00	80.95	19.05

(a) Mistake finding accuracy for traces that do not contain mistakes (correct$_{mis}$).

Model	Direct (trace)	Direct (step)	CoT (step)
Word sorting			
GPT-4-Turbo	32.71	32.33	–
GPT-4	28.20	39.47	30.83
GPT-3.5-Turbo	0.00	4.51	15.04
Gemini Pro	6.39	–	–
PaLM 2 Unicorn	0.38	9.02	11.28
Tracking shuffled objects			
GPT-4-Turbo	31.54	59.23	–
GPT-4	59.14	62.69	92.31
GPT-3.5-Turbo	1.17	1.92	21.92
Gemini Pro	41.54	–	–
PaLM 2 Unicorn	5.38	19.23	56.92
Logical deduction			
GPT-4-Turbo	20.81	74.83	–
GPT-4	39.46	67.01	10.54
GPT-3.5-Turbo	0.00	24.83	7.82
Gemini Pro	8.16	–	–
PaLM 2 Unicorn	4.76	36.73	11.22
Multistep arithmetic			
GPT-4-Turbo	34.27	44.12	–
GPT-4	41.60	41.60	44.54
GPT-3.5-Turbo	0.00	12.18	16.81
Gemini Pro	5.46	–	–
PaLM 2 Unicorn	5.88	2.94	22.27
Dyck languages			
GPT-4-Turbo	6.99	28.46	–
GPT-4	7.37	40.81	43.91
GPT-3.5-Turbo	1.28	6.05	2.08
Gemini Pro	2.25	–	–
PaLM 2 Unicorn	0.38	6.43	17.77

(b) Mistake finding accuracy for traces that contain mistakes (incorrect$_{mis}$).

Table 9: Mistake finding accuracy across 5 tasks for correct$_{mis}$ and incorrect$_{mis}$ traces. The combined scores of Table 9a and Table 9b make up Table 4.